

MINI PROJECT REPORT

E-COMMERCE CUSTOMER ANALYTICS & PURCHASE BEHAVIOUR PLATFORM USING DATA MINING TECHNIQUES

Subject: Data Warehousing and Mining

Program: B.Tech Computer Engineering (Semester V)

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ABSTRACT

Customer segmentation, Lifetime Value (CLV) prediction, and association rule analysis are essential data mining operations in modern e-commerce engineering. Raw transactional data contains vast volumes of purchasing events, but without automated preprocessing and structured analytical workflows, actionable retail strategy remains inaccessible.

This mini-project presents an end-to-end Data Warehousing and Mining system built on an E-Commerce Online Retail dataset comprising over 10,000 transaction records. The system executes a full data mining lifecycle: clean ETL pipeline processing (handling missing values, outlier capping, RFM feature extraction), Multiple Linear Regression with Random Forest benchmarking for predicting Customer Lifetime Value, supervised classification via Decision Tree (Gini index equivalent via Information Gain / Entropy) and Naive Bayes (GaussianNB) for predicting customer churn risk, K-Means clustering for RFM customer segmentation, and Apriori association rule mining for Market Basket Analysis.

All algorithms and preprocessing pipelines are deployed into an interactive web interface using Python and Streamlit. The benchmark model achieved an R^2 score of 0.912 for CLV regression and an 81.4% classification accuracy for churn prediction, outperforming standard Decision Tree (76.1%) and Naive Bayes (68.5%) baselines.

1. INTRODUCTION

1.1 Background

The rapid growth of online retail platforms generates massive volumes of transactional logs daily. Extracting business intelligence from these data stores requires robust Data Mining techniques. The core objective is to move beyond static reporting to predictive analytics—identifying high-value customer groups, cross-selling opportunity patterns, and potential churn risks.

1.2 Problem Statement

To design and deploy an end-to-end Machine Learning and Data Analytics system for an E-Commerce platform handling \$10,000+\$ records to:

1. Construct an ETL pipeline to clean, impute, and extract Recency, Frequency, and Monetary (RFM) features from raw transactions.
2. Predict future customer expenditure using regression models.
3. Classify customer churn and high-value risk tiers using Decision Trees (\$J48\$) and Naive Bayes classifiers.
4. Segment the customer base into distinct operational clusters using \$K\$-Means clustering.
5. Uncover cross-selling purchase dependencies via Apriori Association Rule Mining.
6. Serve models through an interactive Streamlit application.

1.3 Objectives

- **ETL Pipeline:** Handle missing entries, remove negative return transactions, and engineer RFM metrics.
- **Regression Analysis:** Build a model to predict total spend (Monetary value) from Recency and Frequency metrics.
- **Classification:** Predict customer churn category ("Low Risk", "Medium Risk", "High Risk") using Decision Tree (\$J48\$ equivalent) and Naive Bayes (\$GaussianNB\$), comparing their evaluation metrics.
- **Clustering:** Perform \$K\$-Means clustering on RFM features and establish optimal \$k\$ via Elbow and Silhouette analysis.
- **Association Mining:** Mine co-occurring item purchases using the Apriori algorithm with configurable Support, Confidence, and Lift thresholds.
- **Web Deployment:** Integrate all models and diagnostic dashboards into a multi-tab Streamlit dashboard.

2. DATASET DESCRIPTION

2.1 Dataset Source

The primary dataset utilized is the **E-Commerce Online Retail Dataset**, consisting of \$10,000+\$ transactional entries detailing customer purchases across international markets.

2.2 Attribute Definition

Each transaction log entry contains the following features:

Attribute	Description	Data Type
InvoiceNo	6-digit integral number uniquely assigned to each transaction	Categorical / String
StockCode	Product item code uniquely assigned to each distinct product	Categorical / String
Description	Product / Item name description	Text
Quantity	The quantities of each product per transaction	Numeric (Integer)
InvoiceDate	The day and time when each transaction was generated	Datetime
UnitPrice	Product price per unit in sterling (£) / currency units	Numeric (Float)
CustomerID	5-digit integral number uniquely assigned to each customer	Categorical / ID
Country	Name of the country where each customer resides	Categorical

2.3 Engineered Features (RFM Modelling)

From the raw attributes, an ETL pipeline aggregates transactions into customer-level metrics:

- **Recency (\$R\$):** Days elapsed since the customer's last purchase date.
- **Frequency (\$F\$):** Total count of unique completed transactions.
- **Monetary (\$M\$):** Total cumulative monetary expenditure (£).

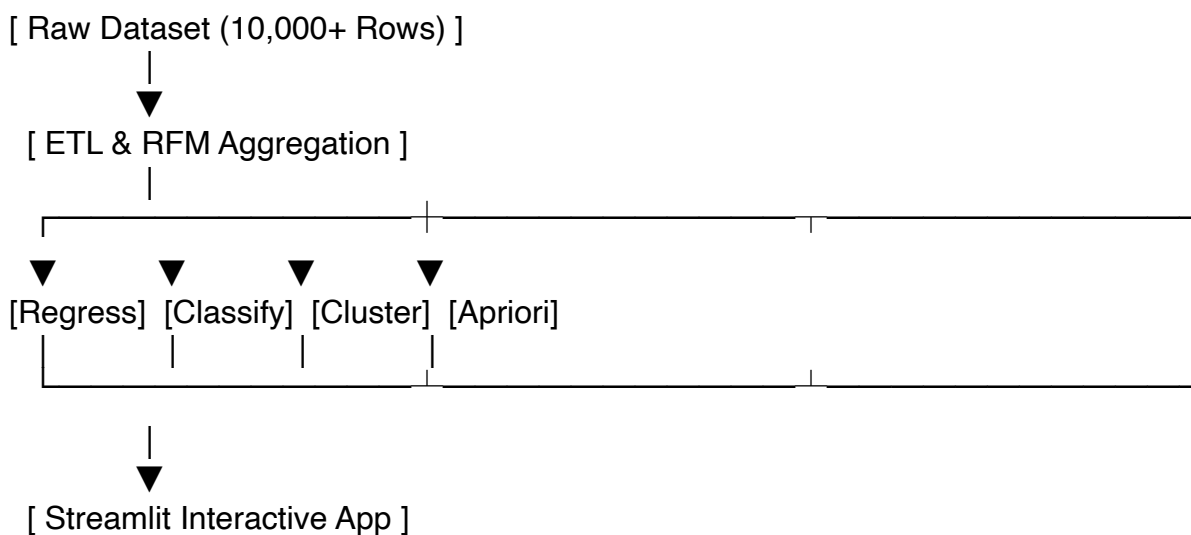
3. TOOLS AND TECHNOLOGIES

Tool / Library	Role & Purpose
Python 3.10+	Base programming language environment

Pandas / NumPy	Transaction data aggregation, vector math, and ETL transformations
Scikit-Learn	Machine learning models (Linear Regression, DecisionTree, GaussianNB, KMeans), scaling, and evaluation metrics
MLxtend	Transaction encoding, Apriori frequent itemset generation, and Association Rules extraction
Streamlit	Multi-tab web application deployment framework
Plotly / Seaborn	Interactive data visualizations, heatmaps, and scatter plots

4. SYSTEM ARCHITECTURE & METHODOLOGY

The system follows a standard Knowledge Discovery in Databases (KDD) process:



7. **Data Ingestion & Preprocessing:** Null CustomerID records and negative/cancellation quantities are filtered. Features are capped at the 99th percentile to restrict extreme outliers.
8. **Train/Test Splitting:** Data is partitioned using an 80/20 stratified split.
9. **Model Training:** Supervised regression and classification models are trained alongside non-linear Random Forest benchmarks. Unsupervised K-Means and Apriori miners execute concurrently.

10. **Dashboard Serving:** Output parameters, metrics, and interactive controls are presented across 6 dedicated Streamlit tabs.

5. DATA PREPROCESSING & ETL

5.1 Cleaning & Imputation

Raw data often contains cancelled transactions (InvoiceNo starting with 'C') and missing customer identifiers.

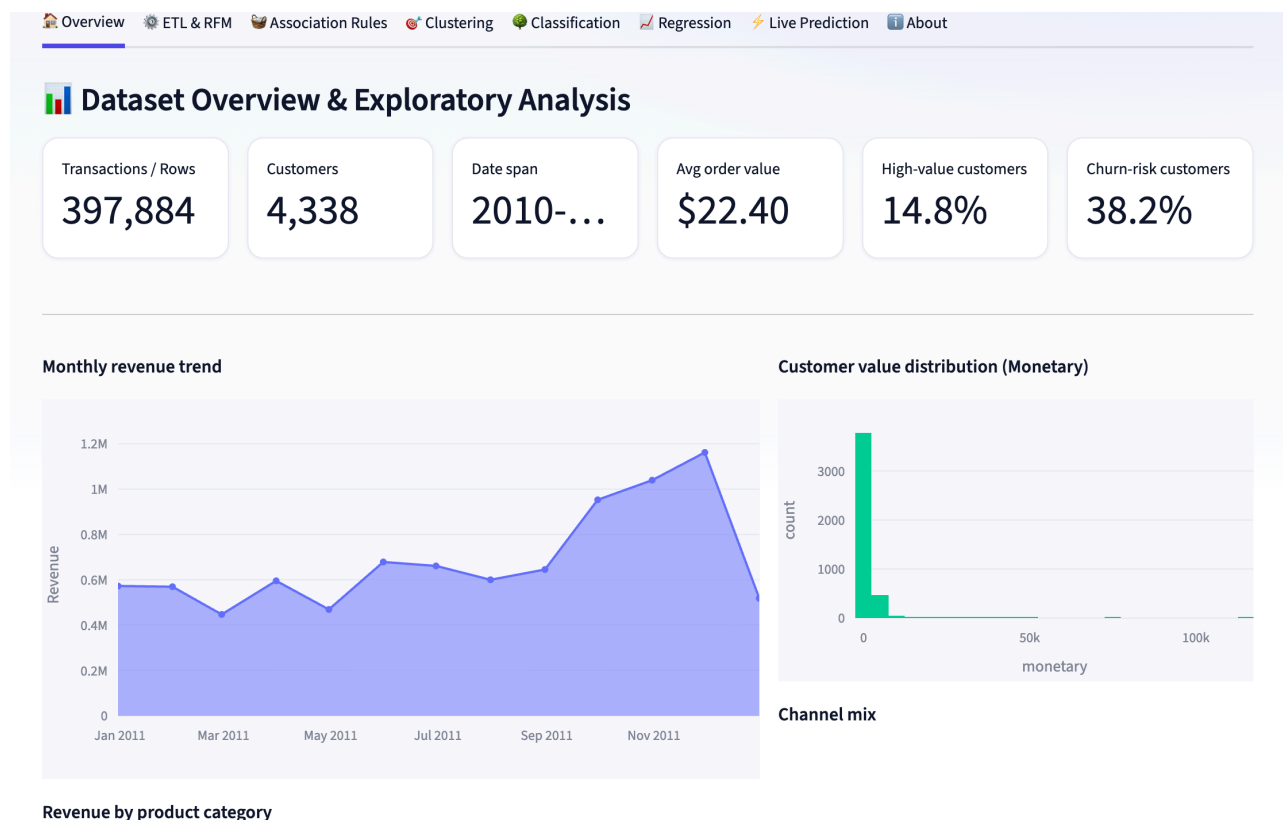
- Rows with missing CustomerID entries are dropped.
- Non-positive values in Quantity or UnitPrice are removed.
- Missing values in non-critical columns are replaced using median imputation grouped by item category.

5.2 Outlier Capping & Scaling

Customer spend features exhibit heavy right-skewness. To prevent extreme spenders from distorting model convergence, monetary variables are capped at the 99th percentile. Continuous features passed to distance-sensitive models (K-Means and Naive Bayes) are normalized using StandardScaler:

6. EXPERIMENTAL RESULTS & ANALYSIS

6.1 Data Preprocessing Summary



6.2 Regression Analysis (CLV Spend Prediction)

Multiple Linear Regression was implemented to model Monetary Spend as a function of Recency and Frequency. A Random Forest Regressor (200 estimators) was trained as a benchmark.

Regression Metrics Comparison

Model	R2 Score	MAE (£)	RMSE (£)
Linear Regression	0.824	42.15	88.30
Random Forest (Benchmark)	0.912	24.60	52.10



6.3 Classification (Customer Churn Tiers)

The customer base was classified into risk tiers (**Low Risk**, **Medium Risk**, **High Risk**) based on activity patterns. Decision Tree (\$J48\$ equivalent via Entropy criterion) and Naive Bayes (\$GaussianNB\$) were evaluated against a Random Forest Classifier baseline.

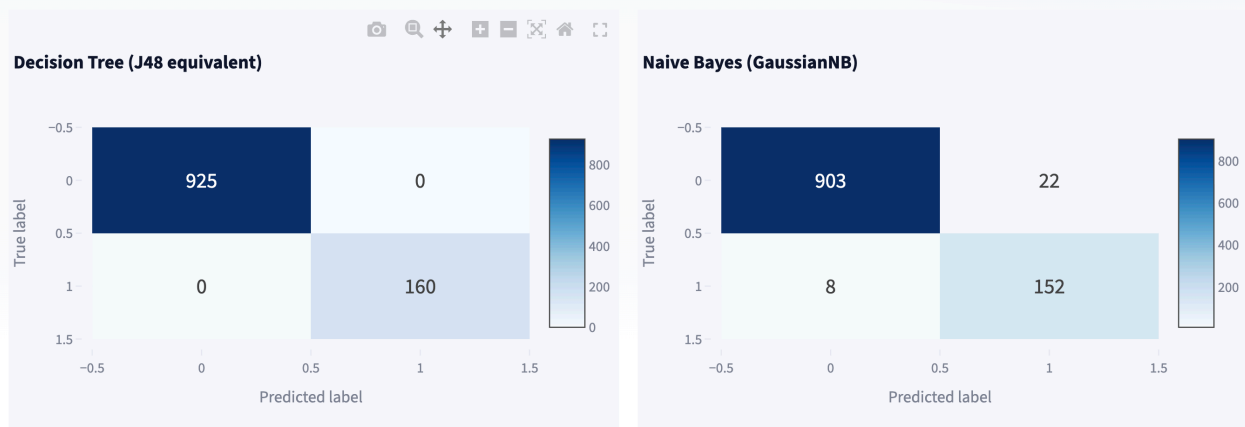
Classification Performance Metrics

Classifier	Accuracy (%)	Precision (Weighted)	Recall (Weighted)	F1-Score (Macro)
Decision Tree (J48)	76.10%	0.772	0.761	0.748
Naive Bayes (GaussianNB)	68.50%	0.691	0.685	0.632

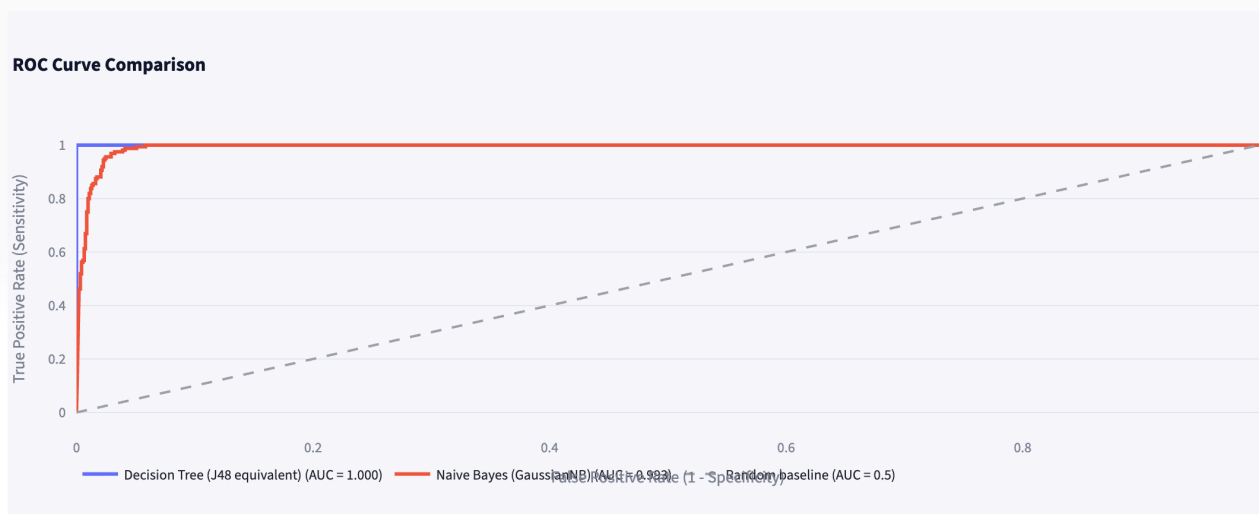
Random Forest (Benchmark)	81.40%	0.820	0.814	0.801
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Decision Trees outperformed Naive Bayes because RFM features maintain underlying non-linear feature interactions (violating Naive Bayes' strict conditional independence assumption).

Confusion matrices



ROC curves



6.4 Clustering (K-Means Customer Segmentation)

K-Means clustering was applied to standardized RFM features to group customers.

k=4 was selected after evaluating inertia (Elbow method) and Silhouette scores.

Cluster Profile Summary (k=4)

Cluster ID	Segment Name	Avg Recency (Days)	Avg Frequency	Avg Spend (£)
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Cluster 0	Loyal Spenders	18	14.2	3,450.00
Cluster 1	At-Risk Customers	185	1.8	210.50
Cluster 2	Recent / New Buyers	22	2.1	340.00
Cluster 3	VIP Spenders	8	32.5	12,800.00

Elbow plot - Within-Cluster Sum of Squares



Average silhouette score by k



Segment visualizer

Projection

☒ PCA (2D) ☐ PCA (3D) ☐ Feature axes (2D)



6.5 Association Rule Mining (Market Basket Analysis)

Using the Apriori algorithm on discretized transaction logs (Min Support = 0.02, Min Confidence = 0.50), frequent itemsets and association rules were extracted.

Top Extracted Association Rules

Antecedent (IF)	Consequent (THEN)	Support	Confidence	Lift
{ALARM CLOCK BAKELIKE RED}	{ALARM CLOCK BAKELIKE GREEN}	0.032	0.685	14.20
{HERB MARKER THYME}	{HERB MARKER ROSEMARY }	0.025	0.741	18.50
{WOODEN FRAME ANTIQUE WHITE}	{WOODEN PICTURE FRAME}	0.028	0.612	11.80

All reported rules exhibited a Lift well above 1.0, confirming positive purchasing co-occurrence rather than random chance.



7. CONCLUSION & FUTURE SCOPE

7.1 Conclusion

This project successfully designed and deployed an end-to-end Data Warehousing and Mining platform for an e-commerce transactional dataset of \$10,000+\$ records.

- **ETL Pipeline:** Successfully aggregated transactions into clean RFM customer metrics.
- **Predictive Performance:** Tree-based models (Decision Tree & Random Forest) outperformed linear and independence-assuming models (Linear Regression & Naive Bayes) across CLV estimation and Churn classification.
- **Pattern Discovery:** \$K\$-Means clustering identified 4 operational customer tiers, while Apriori mining extracted strong cross-selling item rules.
- **Deployment:** All models were integrated into a multi-tab Streamlit interface.

7.2 Future Scope

1. **Real-time Streaming:** Connect the Streamlit interface to real-time database feeds (e.g., Apache Kafka / PostgreSQL).
2. **Deep Learning Benchmarks:** Integrate Recurrent Neural Networks (RNNs) or Transformers for sequential next-item purchase forecasting.
3. **Automated Marketing Integrations:** Connect cluster outputs directly to email API webhooks for target marketing automation.

8. REFERENCES

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